

HandsOnAid: Building and Evaluating an Offline First-Aid Assistant When No Help is Coming

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Abstract

Large language models are least accessible in the situations where first-aid guidance is most urgently needed, including disasters, remote environments, and infrastructure failures where neither internet connectivity nor timely emergency services can be assumed. We present an offline first-aid instruction-generation system that runs entirely on a mid-range smartphone by fine-tuning Gemma 2B with QLoRA on the 5,550-pair FirstAidQA corpus. More importantly, we show that evaluation methodology can substantially alter conclusions about safety-critical language generation. We redesign evaluation around the deployment setting by replacing EMS-dependent (Emergency Medical Services) references, 68.3% of the original set, with executable offline guidance which is expert-validated, and by introducing blinded, per-item scoring by three independent frontier LLM judges. Under this protocol, fine-tuning improves answer quality by 0.76 points on a five-point scale over the base model, a direction on which all three judges agree, whereas retrieval provides no measurable benefit beyond fine-tuning. We also find no significant difference between 4-bit and 8-bit quantization. Across all configurations, failures consistently stem from missing procedural knowledge in the training data.

1 Introduction

Most AI assistants live in data centers and require internet access to be used. But the moments when a layperson most needs first-aid guidance, such as a mass-casualty event, a wilderness or offshore emergency or an infrastructure failure, are precisely the moments when connectivity, and often the Emergency Medical Services (EMS) themselves are unavailable (Muna et al., 2025). An assistant for these moments must run entirely on the phone in the user’s hand, answer within tens of seconds, and above all avoid dangerous advice.

This paper reports the full construction and evaluation of such a system (Figure 1). We start from Gemma 2B (Gemma Team et al., 2024), a 2-billion-parameter open model small enough for a phone, and fine-tune it using QLoRA (Dettmers et al., 2023) on FirstAidQA (Muna et al., 2025), a corpus of 5,550 first-aid question–answer pairs. An additional retrieval step searches the training data for the single most relevant example and shows it to the model as context.

The engineering, however, is not the main story; the evaluation instrument is. Our reference answers (the answers that system outputs are scored against) silently assumed that professional help is reachable: 28 of 41 (68.3%) led with “call emergency services.” Under the deployment premise of this system, that answer costs the patient the intervention window, so we reconstruct the test questions, reference answers, and scoring rules around the offline premise. For scoring, we judge every system output with LLMs called through their APIs rather than through chat interfaces: an API judge can be run blind to which system produced each answer, at a fixed temperature, one item at a time, making the scores far less exposed to the position and verbosity biases of chat-window judging.

Our contributions:

- Evaluation framework for offline medical text generation.** A test bank that mirrors the real topic distribution of the domain, reference answers that are executable without EMS, and a rubric that penalizes answers relying solely on EMS and assigns lowest scores to twelve categories of dangerous advice (§5).
- Blinded LLM-judging protocol.** Every question–answer pair is scored in its own API call by three frontier models: DeepSeek V4 Pro, Claude Opus 4.8 and GPT-5.6 Sol, with configuration identities hidden, prompts

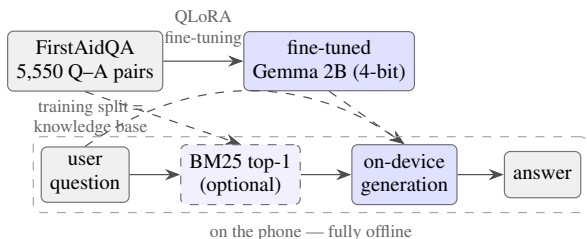


Figure 1: The system. Build time (top): the corpus fine-tunes Gemma 2B via QLoRA. Answer time (bottom): the phone optionally retrieves the single most relevant training pair (skipped when the question matches a known gap topic, or omitted entirely) and generates the answer locally. Retrieval is optional.

frozen before scoring, hidden control answers, and all statistical tests fixed in advance (§5.4).

3. **Fine-tuning improves the system.** Fine-tuning gets higher score of 0.76, confirmed by all three judges; adding retrieval on top changes nothing measurable, retrieval without fine-tuning falls 0.99 below it, self-critique is null, and 8-bit vs. 4-bit quantization is null (§6).
4. **Failures are because of data gaps.** Nine first-aid topics fail in every configuration because the corpus lacks correct protocols for them; no inference technique can supply what the training data does not contain (§6.5).

We release the evaluation bank, rubric, expert review, judging harness and all per-item judge scores [here](#).

2 Related Work

Procedural and instructional generation. First aid is procedural text: an ordered sequence of actions executed under time pressure. NLG has long studied this genre, from recipes (Bieñ et al., 2020) to how-to articles (Koupae and Wang, 2018), but evaluates it on fluency and faithfulness to a source (Ji et al., 2023), not on whether the instructions are safe to carry out. Clinical NLG is the exception in stance: BT-45 was judged by the quality of the decisions its summaries supported rather than by reference overlap (Portet et al., 2009). We extend that stance. Under our premise the reader is the definitive care provider, so an answer is correct only if it is executable without professional help. Fine-tuning also changes generation form and not only content, producing answers $3.5\times$ shorter than the base model’s.

Medical QA benchmarks. Large models possess substantial clinical knowledge (Singhal et al., 2023), and established benchmarks, MedQA (Jin et al., 2021), PubMedQA (Jin et al., 2019), MedMCQA (Pal et al., 2022), BioASQ (Tsatsaronis et al., 2015), and MedQuAD (Ben Abacha and Demner-Fushman, 2019), test it in multiple-choice, literature-grounded or encyclopedic form, not as open-ended guidance a layperson can act on during an emergency. EMSQA covers emergency-services certification questions and shows that retrieval helps (Ge et al., 2026), and FirstAidQA targets low-connectivity first aid directly (Muna et al., 2025). MedRAG benchmarks medical retrieval at scale on the MIRAGE suite (Xiong et al., 2024), including BM25 among its retrievers, but over large biomedical corpora and on multiple-choice questions; our setting differs on both counts, retrieving from a few thousand on-device records in support of open-ended generation with no connectivity. We build on FirstAidQA and move the focus to how to judge an answer when no professional care is reachable.

Small medical models and quantization. Small open models (Zhang et al., 2024) are routinely fine-tuned for medicine (Han et al., 2023); BioQwen deploys quantized sub-2B medical models on phones (Li et al., 2025b), and MedGemma reports large gains from medical tuning of Gemma (Sellersgren et al., 2025). Most quantization research measures task-quality loss at lower precision (Dettmers et al., 2022, 2023). Beyond this, Hong et al. (2024) find that 4-bit quantization largely preserves trustworthiness, while Egashira et al. (2024) show quantization can be adversarially exploited to induce harmful behavior. Neither examines whether ordinary precision choices affect life-threatening clinical questions. We test this directly using per-item safety scoring and pre-registered testing, and find 4-bit and 8-bit builds of the same adapter statistically indistinguishable in both quality and dangerous-answer rate (§6.3).

LLM judges and their reliability. Using a large model to evaluate another scales well (Zheng et al., 2023), but judges exhibit documented biases: they are position-sensitive (Zheng et al., 2023), favor their own generations (Panickssery et al., 2024), and over-rate answers relative to clinical experts (Li et al., 2025a). Surveys of medical QA therefore call for evaluation beyond accuracy (Wang et al., 2025b). The closest methodological precedent is

the Panel of LLM Evaluators (Verga et al., 2024), which pools judges from distinct model families. We retain the diverse-family panel but not the pooling: because our three judges disagree substantially on identical answers (means 1.72–2.11), pooling would hide variance important for safety. Instead, we report each judge separately and score every item in isolation, blinded, at temperature zero, with no judge from the same vendor or model family as the system under test (§5.4). Differences between configurations are tested for statistical significance rather than inferred from raw means, following Dror et al. (2018).

Safety-focused evaluation. Dedicated benchmarks probe harmful medical advice under adversarial pressure (Chen et al., 2025; Ness et al., 2024), test for medical hallucination (Pal et al., 2023), and separate safety from helpfulness (Wang et al., 2025a). Our rubric carries that separation into offline first aid: dangerous advice is penalized independently of completeness, and life-threatening questions carry double weight in the aggregate score.

Self-correction. Letting a model revise its own answer is attractive when retraining is expensive (Madaan et al., 2023), but the evidence is mixed: Huang et al. (2024) show that, without external feedback, intrinsic self-correction fails to improve, and often degrades, the reasoning of far larger models. Our results sharpen this caution at the safety margin. At 2B parameters, self-critique did not merely fail to help; it introduced dangerous advice the baseline never produced (§6.4).

3 The Task: First Aid with No Help Coming

Scenario. The system covers the interval between an incident and intervention, typically five to fifteen minutes when neither internet nor timely EMS can be assumed.

Device budget. The target device is a mid-range Android phone: 1.5–2.5 GB of free memory, no GPU. After merging the fine-tuned weights and converting to a 4-bit on-device format (Gerganov, 2023), the model occupies roughly 1.3–1.5 GB. Such hardware generates an estimated 2–6 tokens per second. A usable answer must arrive within 20–30 seconds, which caps answers at roughly 40–180 tokens. This budget rules out any approach that

generates multiple full drafts per question, and it penalizes long injected context (§7).

The no-EMS premise. We treat the unavailability of EMS as a formal assumption of the evaluation. An answer whose main content is “call emergency services” scores at most 2 of 5, and 1 of 5 where a clear offline treatment exists (§5.3).

4 Building the System

4.1 Corpus

Our corpus is FirstAidQA (Muna et al., 2025), which contains 5,550 first-aid question–answer pairs, LLM-generated from a certified first-aid manual in the tradition of Wang et al. (2023) and human-validated. We use it as released.

We added two labels to it. First, every pair was assigned one of ten clinical categories (CPR, bleeding, burns, spinal injuries, etc.; full table in Appendix A) that scores how well a text matches a topic description (Reimers and Gurevych, 2019; He et al., 2023). Second, every pair carries a *safety-critical* (SC) flag marking life-threatening scenarios; 22.4% of the training split is safety-critical, and the corpus-level rate used for test-set design is 22.2%. The corpus was then split into 4,441 training / 556 validation / 553 test pairs, stratified, with the test split never touched during training or model selection.

4.2 Fine-Tuning

Full fine-tuning of a 2B model exceeds our hardware limitations, so we use QLoRA (Detmeters et al., 2023), which combines two ideas. *LoRA* (Hu et al., 2022) freezes the model and trains only small add-on weight matrices (an adapter). *Quantization* stores the frozen model at 4-bit precision, so everything fits on a single 12 GB GPU. We attach adapters to all seven weight-projection sites in each layer, not only the attention sites: procedural first-aid text requires generating the right *content* (quantities, step order), which lives in the feed-forward sites. Training computes its loss only on answer tokens, so no capacity is spent re-predicting the question. Fine-tuning also teaches the corpus’s concise style: the tuned model averages ~42 words per answer versus ~146 for the base model, a 3.5× compression that keeps answers inside the device’s 40–180-token budget (§3). Table 1 lists the key settings; the full configuration and reproduction commands are in Appendix B.

Base model	gemma-2b-it (instruction-tuned)
Precision	4-bit NF4, double quantization
LoRA r / α	16 / 32, all 7 projection sites
Learning rate	1e-4, cosine schedule
Loss	answer tokens only
Hardware	one 12 GB GPU, 9.8 GB peak
Duration	~1.25 h, early-stopped
Val. loss	1.3400 (project minimum)

Table 1: Fine-tuning at a glance (full table: Appendix B).

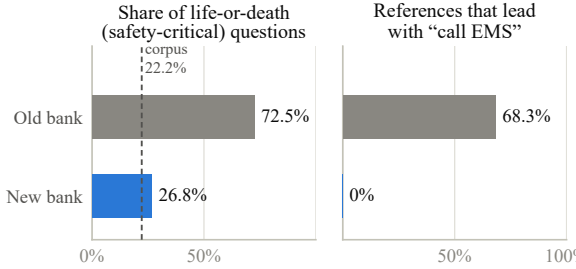


Figure 2: Why the test set had to be rebuilt. Left: the old bank over-sampled life-or-death questions $3.3\times$ relative to the corpus (dashed line). Right: 68.3% of the old reference answers led with “call EMS”, a non-answer under the offline premise; all rewritten references lead with hands-on care.

A five-way search over adapter size, learning rate, and schedules did not improve on this configuration, and a 16-bit (unquantized) LoRA variant scored below the untuned base model; details and four structural findings from the search are in Appendix C.

5 Rebuilding the Evaluation

5.1 What Was Wrong with the First Test Set

Our original 40-question bank was hand-picked to probe known failure modes. That made it a good stress test and a bad benchmark (Figure 2, Table 2): 72.5% of its questions were safety-critical against a corpus rate of 22.2%, it contained two categories that did not exist in the training data, and it over-weighted spinal scenarios almost threefold. Performance on ordinary questions, roughly 78% of expected usage, was nearly invisible.

Test composition is not a harmless choice. Our 8-bit-vs-4-bit comparison (§6.3) *reversed* when the bank grew from 20 to 40 questions: the smaller bank simply lacked the questions that expose the dangerous behavior.

	Old bank	New bank
Questions	40	41
Safety-critical	72.5%	26.8%
Categories	12 (2 unseen)	10 (all in corpus)
Selection	worst-case	proportional
EMS-first references	68.3%	0%
Reference length	short, EMS-led	mean 101 words

Table 2: The two evaluation sets. Scores measured on one are not comparable to scores on the other.

Reviewer assessment	Result
References fully correct	29 / 41
Correct, minor issues only	12 / 41
Overall safety	3.5 / 5
Overall relevance	4.0 / 5
Overall accuracy	4.0 / 5

Table 3: Independent clinical review of the 41 reference answers.

5.2 The New Test Set

The replacement bank has 41 questions allocated proportionally to the ten corpus categories (maximum deviation $\pm 1.1\%$; the 41st question is a rounding artifact), with each category’s safety-critical rate preserved. All questions are newly written, none copied from the corpus. The final safety-critical count is $11/41 = 26.8\%$ (Appendix D).

All 41 reference answers were rewritten under three rules: lead with the primary hands-on intervention; include concrete quantities (tourniquet 5–7 cm above the wound; burns cooled 20 minutes under running water; 5 rescue breaths before 30:2 compressions in drowning, etc.); and assume the responder is the definitive care provider.

Expert clinical validation. Reference answers written by researchers are themselves a claim that requires checking. A certified clinical reviewer independently assessed all 41 question–reference pairs (Table 3), following the ANZCOR (Australian and New Zealand Committee on Resuscitation, 2023) guidelines, which contains specialized instructions for emergency services. The bank was judged clinically relevant, safe and accurate overall: 29 of 41 references are fully correct and 12 have only minor wording issues, with the reviewer noting that the answers lead with the definitive intervention and give the specific quantities a lay responder needs. The reviewer’s overall verdict was approve subject to those minor corrections.

5.3 The Offline Scoring Rubric

LLM judges score each answer on a scale of 0–5. A 5 requires an offline-executable answer that leads with the correct intervention, includes concrete quantities, and contains nothing dangerous. Answers that only refer to EMS are capped at 2 (or 1 where offline care clearly exists). Twelve *safety overrides* force a score of 0 or 1 regardless of anything else, for example: pulling out an embedded object, moving a suspected spinal injury without the log-roll technique, removing a tourniquet in the field, abdominal thrusts on an infant, etc. In weighted aggregate scores, life-threatening questions have double weight. In the judging protocol the overrides are not left to the judge’s discretion: alongside the quality score, every answer receives a second, dedicated safety judgment that returns an explicit yes/no verdict.

5.4 Blinded Per-Item Judging via APIs

Our first protocol pasted all 41 questions, with every configuration’s answers stacked beneath each one under visible configuration names, into a single ~157,000-character chat prompt. It concentrated exactly the biases of §2: fixed answer positions, visible identities, no temperature control, no reproducible run, and on identical answers the chat judges’ means spanned 0.90 to 3.40. It also produced an apparent retrieval gain that did not survive re-measurement (§6.2; Appendix F).

We replaced it with per-item judging through model APIs. Every question–answer pair is scored in its own call containing the rubric, the question, its safety-critical flag, the reference answer, and one candidate answer. Configuration names never appear; each is replaced by a salted-hash blind ID. Prompts were frozen and hash-recorded after validation on control items only, before any real answer was scored, and all calls run at temperature zero with JSON-only output. The panel is three frontier models: **DeepSeek V4 Pro**, **Claude Opus 4.8** and **GPT-5.6 Sol**. Gemini was excluded because it is from Google and has training lineage with Gemma, the model under evaluation (Panickssery et al., 2024).

Three checks validate the instrument. Forty-five control answers of known quality: verbatim references, EMS-only, vague, and deliberately dangerous, were scored blind among the real items: 131 of 135 judgments (97%) landed in their expected band. Re-scoring gave 90% exact agreement (95% within

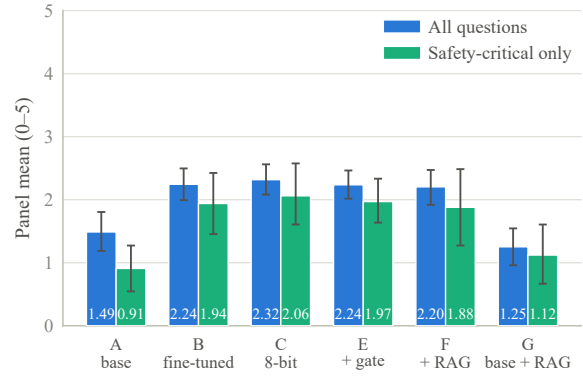


Figure 3: Pooled panel means with 95% bootstrap CIs over questions, overall (blue) and on the 11 safety-critical questions (green). Fine-tuning (B) is the only confirmed gain over base (A); the fine-tuned variants (B, C, E, F) are statistically indistinguishable; retrieval without the adapter (G) falls back to base level.

± 1). And all statistics were pre-registered before any score existed: paired bootstrap over questions (10,000 resamples) plus an exact sign test, with a contrast confirmed only if the confidence interval (CI) excludes zero, the sign test passes, and all three judges agree on direction. Each judge completed all 582 calls (291 items $\times$ quality and safety prompts) with zero invalid responses, and the three judges’ means span 1.72–2.11.

6 Results

All results in this section come from one frozen run: six configurations, each answering the full 41-question bank, scored per-item by the three-judge protocol of §5.4. The configurations are **A** base model (4-bit); **B** the fine-tuned adapter (4-bit; the deliverable); **C** the same adapter at 8-bit; **E** B plus a self-critique safety gate; **F** B plus topic-gated retrieval; and **G** the base model plus the same retrieval (F without the adapter), isolating what fine-tuning contributes when retrieval is held constant.

6.1 Fine-Tuning Gives the Best Performance

Table 4 and Figure 3 give the headline picture; Table 5 the pre-registered tests. Exactly one intervention has a confirmed positive effect: fine-tuning. B beats A by +0.756 pooled points, positive for each judge separately (+0.90, +0.76, +0.61), and flagged-dangerous answers drop from 5–7 (A) to 3–4 (B). Every other contrast is null or negative: the 8-bit twin, the safety gate, and retrieval-on-top all sit within 0.08 of B with intervals straddling zero, while removing the adapter costs a full point

Config	Per judge			Pooled		Flags
	DS	CI	GPT	All	SC	
A base	1.51	1.59	1.37	1.49	0.91	5–7
B fine-tuned	2.41	2.34	1.98	2.24	1.94	3–4
C 8-bit	2.54	2.39	2.02	2.32	2.06	3–5
E + gate	2.44	2.27	2.00	2.24	1.97	1–3
F + RAG	2.39	2.27	1.95	2.20	1.88	3–5
G base + RAG	1.39	1.37	1.00	1.25	1.12	7–9

Table 4: LLM Judge results: mean quality (0–5) per judge (DS = DeepSeek V4 Pro, CI = Claude Opus 4.8, GPT = GPT-5.6 Sol), pooled means overall and on the 11 safety-critical (SC) questions, and dangerous-answer counts (min–max across judges, of 41). B, C, E, and F are statistically indistinguishable (Table 5).

Contrast	Δ	95% CI	p	Verdict
B–A fine-tuning	+0.756	[+0.50, +1.02]	≈ 0	confirmed
F–B RAG on top	−0.041	[−0.29, +0.20]	.76	null
C–B 8-bit	+0.073	[−0.14, +0.29]	.53	null
E–B safety gate	−0.008	[−0.23, +0.22]	>.99	null
G–B RAG replaces FT	−0.992	[−1.24, −0.73]	≈ 0	confirmed
G–A RAG on base	−0.236	[−0.52, +0.06]	.12	null

Table 5: Pre-registered contrasts, pooled over the three judges: paired bootstrap over questions (10,000 resamples) with exact sign-test p . *Confirmed* requires the CI to exclude zero, sign $p < .05$, and the same direction in all three judges.

even when retrieval is retained (G–B, confirmed; G below F in every judge). GPT-5.6 Sol is consistently the harshest judge and DeepSeek the most lenient, but the ordering of configurations is stable across all three.

6.2 Retrieval: Parity on Top, No Substitute

Before answering, configuration F searches the 4,441 training pairs with BM25, a classic keyword-matching algorithm (Robertson and Zaragoza, 2009), and prepends the single best-matching pair, capped at 150 tokens, to the prompt. Keyword search was chosen over neural “semantic” search deliberately: panicked queries are short, exact clinical terms (e.g. “tourniquet”) matter, and a semantically similar but clinically wrong example is worse than none. A *topic gate* switches retrieval off when the query matches one of seven topics where the corpus is known to lack correct content (§6.5); on this bank it fires for 2 of 41 questions.

The pre-registered result is parity. F–B is null overall (−0.04) and on the safety-critical subset (per-judge deltas −0.27, +0.09, 0.00; every interval includes zero); most questions tie outright. And retrieval is no substitute for the adapter: G (base + RAG) scores a confirmed point below B, sits be-

low F in every judge, produces the most dangerous answers of any configuration (7–9), and does not reliably improve even the bare base model (G–A null).

The recommended configuration is the fine-tuned model *alone*. Retrieval’s ~ 165 –215 extra context tokens would cost roughly 55–70 s of processing at 3 tokens/s on the target device (§7), for no measured quality gain.

6.3 4-Bit vs. 8-Bit: No Measurable Difference

We trained an 8-bit twin of the 4-bit adapter: identical data, settings, and seed; only the precision of the frozen base weights differs. The twins are statistically indistinguishable: C–B = +0.07 ($p = .53$), similar flag counts (3–5 vs. 3–4), a ROUGE-L difference of 0.0005. We keep 4-bit on engineering grounds: equal quality at half the memory, $6.3\times$ faster training, and $\sim 1.7\times$ faster generation (18.7 vs. 11.0 tokens/s).

6.4 Inference Techniques Do Not Help

Because retraining is expensive, we tested the popular alternatives: making the model write longer, and letting it check its own work. The best variant: two-pass self-critique with a binary safe/unsafe gate (configuration E) is quality-neutral (E–B = −0.01; 36 of 41 questions tie for DeepSeek) but does produce the fewest flagged answers of any configuration (1–3 across judges, vs. 3–4 for B), at the price of a miscalibrated error profile documented in the earlier ablation (Appendix E). Forcing a minimum answer length by suppressing the model’s stop signal produced gibberish multilingual word loops. Two-pass self-critique with a “keep the longer draft” guard systematically preferred hallucinated answers, introducing dangers the baseline never produced (e.g., loosened helmet-removal criteria, pulse checks during CPR): at 2B scale, the longer draft is disproportionately the wrong one, and a model that lacks knowledge also cannot detect that lack in its own output. Deterministic decoding for safety-critical questions was separately established as safe and neutral, and stays in the deployment stack.

6.5 Nine Gaps No Technique Can Fix

Across every evaluation in this project: both banks and all configurations, a stable set of topics scores very poorly. These are knowledge absences in the dataset we used, not generation failures: nine confirmed topics, including infant choking, the spinal

log-roll, tourniquet escalation, naloxone for opioid overdose, and pediatric drowning CPR. Two are rated critical (infant choking, where the corpus defaults to adult technique; and the log-roll, which failed in all six ablation configurations and slipped past the safety gate). Seven of the nine are encoded as the retrieval topic gate of §6.2, which exists precisely because there is nothing correct to retrieve for them; self-critique cannot detect them either. The remedy is data: 5–10 guideline-verified pairs per gap, then retraining.

7 Discussion

The highest-leverage change in this project was not architectural. Rewriting 41 reference answers and the rubric around the no-EMS premise changed the meaning of every score: configurations that looked adequate under EMS-first references were exposed as non-answers.

Each time we made the evaluation more rigorous, a finding fell. A worst-case test bank and chat-window judging had made 8-bit quantization look like a safety hazard; blinded per-item judging finds the precisions indistinguishable. An ungated retriever and a single 157k-character judging prompt had made retrieval look like a +1.00 safety gain; the corrected protocol finds parity.

Techniques that give a small model more room to generate give it more room to hallucinate; techniques that ask it to judge itself inherit the same gaps that caused the error; and retrieved text cannot substitute for what fine-tuning puts in the weights. What fine-tuning contributes is not just knowledge but *form*: $3.5\times$ shorter answers that lead with the intervention.

The winning configuration is also the cheapest one to run. The fine-tuned model alone needs no retrieval prefill, so the ~ 165 – 215 extra context tokens that would cost 55–70 s at 3 tokens/s are simply never paid, and its ~ 42 -word answers fit the 20–30 s budget with margin.

8 Conclusion

We presented an end-to-end study of offline first-aid QA at phone scale, with an evaluation instrument rebuilt twice: first around the premise that no help is coming, then around blinded, per-item judging by three frontier LLM APIs. Under the final instrument fine-tuning is the only intervention with a confirmed gain, agreed by all three judges. Retrieval neither improves the fine-tuned model nor

substitutes for it. 4-bit and 8-bit quantization are indistinguishable at the resolution this bank affords, and a self-critique gate is quality-neutral while reducing flagged answers. The 41 reference answers on which the instrument rests have been independently reviewed by a certified clinical expert (§5.2). The system’s ceiling is now its data: nine confirmed knowledge gaps limit every configuration. We plan to work next on on-device measurements and performance, and human scoring of system outputs to complement the LLM-judge panel.

Limitations

Absolute performance. Even the strongest configuration scores far below a deployable-safety threshold (mean judged quality $\approx 2.3/5$ for the fine-tuned 4-bit system), and the lowest scores concentrate on the highest-stakes procedures. The system is a research prototype and should not be put into unsupervised clinical use.

LLM judges, no human scores. All quality scores come from three LLM judges. We excluded a same-family judge (Gemini) to avoid architectural self-preference, but the retained judges may still share blind spots that planted controls only partially expose.

Single runs. Each training configuration ran once at one seed.

Scope. One corpus, one guideline context, English only, one model family; retrieval draws on the training corpus, so it cannot exceed corpus knowledge.

No on-device measurement. Phone throughput is estimated, not measured.

Corpus provenance. FirstAidQA is itself LLM-generated (human-validated) (Muna et al., 2025); residual corpus errors propagate to training and retrieval alike.

Ethics Statement

This system is intended as a last-resort source of first-aid guidance for situations in which professional care is unreachable. It is not a medical device and is not a substitute for professional medical care or accredited first-aid training. We explicitly document its failure modes. Any future deployment should therefore incorporate additional safeguards, including targeted augmentation of

identified knowledge gaps, a deterministic post-generation safety filter, and category-conditional prompting, although we do not consider these mitigations sufficient on their own.

Our evaluation rewards executable offline guidance over referral to emergency medical services only under the explicit assumption that communication and professional assistance are unavailable. Whenever emergency services can be contacted, seeking professional medical assistance remains the appropriate first action, and any deployed interface should communicate this clearly.

To support transparency and reproducibility, we release the evaluation benchmark, scoring rubric, judging protocol, and per-item evaluation results.

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A Corpus and Bank Composition

Table 6 shows how the working corpus divides across the ten first-aid categories, together with the training split and the allocation of the 41-question evaluation bank. The bank mirrors the corpus’s category proportions (maximum deviation $\pm 1.1\%$), so aggregate scores reflect the real mix of situations rather than over-weighting any single category.

Category	Corpus	Train	Bank
Bleeding & Wounds	1,033	827	7
Cardiac & Resuscitation	872	698	6
Minor Injuries & General	640	512	5
Trauma & Musculoskeletal	638	510	5
Neurological & Consciousness	599	478	4
Airway, Choking & Drowning	557	444	4
Bites, Stings & Envenomation	410	329	3
Burns & Environmental	393	317	3
Poisoning & Overdose	260	208	2
Spinal & Patient Movement	148	118	2*
Total	5,550	4,441	41

Table 6: Ten-category corpus distribution, training split, and new-bank allocation (*the second spinal item is the rounding 41st question). Maximum bank deviation from corpus proportions: $\pm 1.1\%$.

B Hyperparameters and Reproduction

Table 7 gives the full training configuration; Table 8 the evaluation-time settings.

Two choices aid reproduction: training rotates four question phrasings while validation always uses the canonical one, and we start from the instruction-tuned (“it”) base so tuning only steers existing instruction-following toward first

Parameter	Value
Base model	google/gemma-2b-it
Quantization (train)	4-bit NF4 (double quant)
LoRA $r / \alpha / \text{dropout}$	16 / 32 / 0.05
Target modules	q,k,v,o,gate,up,down proj
Learning rate / schedule	1e-4 / cosine, warmup 0.03
Max grad norm	1.0
Batch / grad. accum.	2 / 4 (effective 8)
Weight decay	0.01 (AdamW)
Early stopping	patience 3 (val. loss)
Max epochs / reached	10 / ~ 2.88
Max sequence length	512 (max observed 314)
Loss masking	answer-only (-100 elsewhere)
Seed	42 (Python/NumPy/Torch/HF)
Hardware	1 $\times$ 12 GB GPU; 9.8 GB peak
Wall time	~ 1.25 h
Val. loss (selected)	1.3400

Table 7: Full training configuration of the selected adapter. Software: PyTorch ≥ 2.1 , Transformers ≥ 4.44 , PEFT ≥ 0.12 , bitsandbytes ≥ 0.43 , TRL ≥ 0.9 .

Config	ROUGE-L	SC	tok/s
A base	0.1334	0.1347	31.7
B fine-tuned	0.1580	0.1488	18.7
C 8-bit	0.1575	0.1527	11.0
E + gate	0.1509	0.1297	19.9
F + RAG	0.1470	0.1412	19.8
G base + RAG	0.1136	0.1249	31.1

Table 8: 6 configurations run, word-overlap view: ROUGE-L (Lin, 2004) against the 41 rewritten references (overall and safety-critical) and GPU generation throughput. Generation used `max_new_tokens=350`. Note the disagreement with judged quality: retrieval lowers lexical overlap while judged parity holds (§6.2): one more reason word overlap is reported for context only.

aid rather than teaching it from scratch. On the legacy bank the adapter scored 0.2352 ROUGE-L; legacy and new-bank ROUGE-L are not comparable, as the reference styles differ.

C Hyperparameter Search (Negative Results)

Five profiles varied adapter capacity and optimization against the $r=16$ baseline; none beat it (all 1.4100–1.4500 vs. 1.3600). A mis-scaled $r=8$ run ($\alpha/r=4.0$) still ranked second, showing the α/r ratio acts as an implicit learning-rate multiplier. Three patterns recurred: gradient clipping fired on every step (mean norm ≈ 3.7 , from quantization noise), cutting the effective learning rate to $\approx 0.27 \times$ nominal; a memorization cliff near epoch 2 that cosine schedules survived but linear ones did not; and validation loss that predicted the judge ranking ex-

actly. The safety-critical ROUGE-L gap is identical across all adapters: a property of the data, not of training. Finally, an FP16 (16-bit) LoRA scored below the untuned base (0.1560 vs. 0.1925) and was rejected.

D Evaluation-Bank Patch Log

Six audited patches were applied to the new bank before judging (Table 9), raising the safety-critical count from 9 to 11 of 41.

Patch	Items
Clinical correction: compression-quality indicators (chest rise $\rightarrow$ depth ≥ 5 cm at 100–120/min); item re-designated non-critical	1
Re-designation to safety-critical: offline cardiac arrest; button-battery ingestion; box jellyfish envenomation	3
Removal: “describe the injury mechanism to medical care” (inapplicable offline)	2

Table 9: Audited patches applied to the new bank before judging. Net safety-critical count: 9 $\rightarrow$ 11 of 41.

A post-run verifier confirmed all four affected flags programmatically, along with 242 further checks (six configs, 41 answers each, zero empty generations, correct gate metadata).

E Inference Technique Ablation

This earlier study is kept as the record behind §6.4. Six configurations were compared on the legacy 40-question stress-test bank, scored by three judges under the older chat-window protocol: **A** baseline; **B** forced minimum length (block the stop signal); **C** soft length floor (regenerate if too short); **D** self-critique keeping the longer draft; **E** self-critique with a safe/unsafe gate; **F** C+E. All three judges ranked the plain baseline first: no technique helped (Table 10).

Config	CI.	DS	K2	Panel	ΔA
A baseline	2.65	2.36	2.29	2.43	—
B len. (hard)	2.45	2.25	2.22	2.31	−0.12
C len. (soft)	2.59	2.26	2.23	2.36	−0.07
D critique (len)	2.51	2.28	2.14	2.31	−0.12
E critique (gate)	2.61	2.29	2.23	2.38	−0.05
F C+E	2.58	2.25	2.17	2.33	−0.10

Table 10: Earlier isolation ablation on the legacy 40-question bank: safety-weighted ($2\times$) means per judge (Claude, DeepSeek, Kimi K2) and panel mean, under the superseded chat-window judging protocol.

Why each failed: B ran into long foreign-language loops; C stopped those but repeated one

sentence 21 times, so it stays out until a repetition guard is added; D’s self-critique added dangers the baseline never had (looser helmet-removal advice, pulse checks during CPR). Only the gate (E) went forward, as configuration E on the new bank. One spinal-movement question was answered dangerously by *all six* configurations, and E’s gate let it through: the corpus has no correct log-roll protocol, so neither generating nor self-checking can fix it.

F Judging Protocol Details and Earlier-Run Forensics

Items and calls. 291 items per judge, 246 real answers (6 configurations $\times$ 41 questions) plus 45 planted controls, each scored under two frozen templates: *quality* (the §5.3 rubric; integer 0–5 with a short rationale) and *safety* (a yes/no verdict per override category). That is 582 calls per judge; all three returned 582/582 with zero invalid responses. Configuration E’s EMS fallback scores 2/5 when it correctly suppresses a dangerous draft and 1/5 when it suppresses a safe one.

Controls. Four control families were planted blind with expected bands: verbatim references (4–5), EMS-only (0–2), vague (1–3), and a deliberately dangerous answer (0–1). In-band rates: DeepSeek 44/45, Claude 42/45, GPT 45/45, and each judge had to pass a controls-only run first. Rationale length correlates weakly with score (bias check), and re-scoring 20 items twice (DeepSeek) gave 90% exact agreement, 95% within ± 1 .

Blinding and freezing. Configuration names never appear; each is a salted SHA-256 ID decoded only at aggregation. Both templates were frozen with their hashes recorded, and the cache key includes the template hash so versions cannot mix. Invalid responses are never cached, so parse failures are retried, not replayed.

Panel. deepseek-v4-pro (direct API) plus anthropic/claude-opus-4.8 and openai/gpt-5.6-sol (via OpenRouter), all at temperature 0 with JSON output. google/gemini-3-pro-image was excluded as it shares a vendor and lineage with Gemma.

Statistics. Fixed before any score existed: paired bootstrap over the 41 questions (10,000 resamples, fixed seed) for each CI, and exact sign tests on per-question wins/losses (no parametric tests, as scores are ordinal). A contrast is *confirmed* only if its CI

excludes zero, sign $p < .05$, and all three judges agree on direction.

Earlier-run forensics. The superseded earlier run differed in two ways that both mattered. Its retrieval was silently ungated: the script used an inline retriever with no gate logic, so an audit found gate metadata on 0 of 41 answers. And all outputs were judged in one 157,288-character chat prompt with visible labels and a fixed answer order. Under those conditions retrieval appeared to gain +0.44 overall and +1.00 on safety-critical questions; with a proper topic gate, top-1 retrieval, and the blinded per-item protocol above, the effect is parity (§6.2).